Trials@uspto.gov
571-272-7822

Paper 35
Date: January 26, 2026

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

DR. SQUATCH, LLC,
Petitioner,

v.

THE PROCTER & GAMBLE COMPANY,
Patent Owner.

IPR2024-01105
Patent 10,966,915 B2

Before SHERIDAN K. SNEDDEN, SUSAN L. C. MITCHELL, and CHRISTOPHER M. KAISER, *Administrative Patent Judges.*

SNEDDEN, *Administrative Patent Judge*.

JUDGMENT

Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

Dr. Squatch, LLC ("Petitioner") filed a Petition requesting *inter partes* review of claims 1–15 of U.S. Patent No. 10,966,915 B2 (Ex. 1001, "the '915 patent"). Petition ("Pet."), Paper 2. The Procter & Gamble Company ("Patent Owner") filed a Preliminary Response identifying itself as the owner of the '915 patent. Preliminary Response ("Prelim. Resp."), Paper 7.

We instituted trial on January 23, 2025. Paper 12 ("Inst. Dec."). During trial, Patent Owner filed a Patent Owner Response. Paper 16 ("PO Resp."). Petitioner filed a Reply (Paper 20 ("Reply")) and Patent Owner filed a Sur-reply (Paper 25 ("Sur-reply")). We held an oral hearing on October 22, 2025, and a transcript of that hearing is of record. Paper 34 ("Tr.").

We have jurisdiction under 35 U.S.C. § 6(b). After considering the full record developed through trial, we determine that Petitioner has proved by a preponderance of the evidence that the challenged claims are unpatentable. *See* 35 U.S.C. § 316(e). Our reasoning is explained below, and we issue this Final Written Decision under 35 U.S.C. § 318(a).

II. BACKGROUND

A. Real Parties in Interest

Petitioner identifies Dr. Squatch, LLC as its real party-in-interest. Pet. 1. Patent Owner identifies The Procter & Gamble Company as its real party-in-interest. Paper 4, 1.

B. Related Matters

The parties indicate that the '915 patent is involved in *The Procter & Gamble Company v. Dr. Squatch LLC*, Case No. 2-24-cv-04711 (C.D. Cal.), filed November 29, 2022. Pet. 1; Paper 4, 1. Petitioner has also filed petitions for *inter partes* review of the following patents: IPR2024-01104 against IPR2024-01173 against U.S. Patent No. 10,905,647; IPR2024-01174 against U.S. Patent No. 11,497,706; and IPR2024-01498 against U.S. Patent No. 11,844,752. Paper 4, 1; Paper 33, 2.

C. The ’915 patent (Ex. 1001)

The ’915 patent is titled “Deodorant Compositions.” Ex. 1001, code (54). The ’915 patent is directed to deodorant compositions and methods of making the compositions. *Id.* at 1:15–16, 10:65. More specifically, the ’915 patent is directed to formulating, in response to consumer demands, an aluminum-free, mostly silicone-free deodorant stick, which is not too hard. *Id.* at 1:21–22, 1:32–34.

The ’915 patent explains that

> [c]onsumers also want a good glide, non-sticky, and non-greasy application. This is a challenge, because a mostly silicone-free formula will often use natural oils or natural oil-based triglycerides. But often natural oils, such as coconut oil, bring a large portion of wax with them, often making the resulting deodorant sticks too hard to meet a more preferable product application for both shaven and unshaven underarms. They are also often undesirably greasy due to the non-volatile nature.

Id. at 1:22–30. “As consumers seek more natural ingredients in their deodorants, one approach to formulation is to use emollients derived from natural oils.” *Id.* at 2:59–61. The specification describes that products aimed at emphasizing the use of natural ingredients “may have a significant amount of a liquid triglyceride.” *Id.* at 2:66. To provide structure to these products, the ’915 patent describes that it is known to include waxes and other such structurants. *Id.* at 3:1–3. However, the ’915 patent explains that

> a currently marketed product, Comparative Formula 1 . . . , has 34.15% liquid triglyceride, along with a number of structurants, resulting in a very hard stick, scoring 63 on a needle penetration test under ASTM D-1321 . . . . So while Comparative Formula 1 uses consumer preferred natural ingredients, it does not necessarily provide a good consumer experience when used, given its hardness. Comparative Formula 2 contains even higher levels of structurants, resulting in a harder stick with an even lower hardness score. In comparison, Inventive Examples 1–5, while comprising consumer-preferred natural ingredients, have higher hardness scores, meaning they are softer products.

Id. at 3:5–17. According to the ’915 patent, “a deodorant stick having at least about 25% of a liquid triglyceride, and that uses a primary structurant that has a melting point of at least about 50° C., . . . can result in a deodorant stick with a hardness from about 80 mm*10 to about 140 mm*10.” *Id.* at 3:49–57. The ’915 patent describes that the “deodorant stick is able to comprise consumer-perceived natural ingredients, while offering a pleasant consumer experience in terms of its hardness.” *Id.* at 3:58–60.

The ’915 patent describes that the deodorant compositions “can contain a structurant, an antimicrobial, a perfume, and additional chassis ingredient(s) . . . in the form of a solid stick.” *Id.* at 9:23–27. The ’915 patent defines a “structurant” as “any material known or otherwise effective in providing suspending, gelling, viscosifying, solidifying, and/or thickening properties to the composition or which otherwise provide structure to the final product form.” *Id.* at 5:1–5. The ’915 patent describes a deodorant stick comprising a magnesium hydroxide that is effective against underarm bacteria strains in raw material evaluations. *Id.* at 8:27–35, 8:43–46.

D. *The Challenged Claims*

Petitioner challenges claims 1–15. Representative independent claim 1 is reproduced below:

1. A deodorant stick comprising:

   a. at least about 25% of a liquid triglyceride;

   b. at least one antimicrobial; and

   c. a primary structurant with a melting point of at least about 50° Celsius; and

   wherein said stick is free of an aluminum salt.

Ex. 1001, 15:37–43.

E. Evidence

Petitioner relies upon information that includes the following.

Ex. 1003, Lesniak et al., US 11,433,018 B2, issued September 6, 2022 (“Lesniak”).

Ex. 1004, Phinney et al., US 9,314,412 B2, issued April 19, 2016 (“Phinney”).

Ex. 1005, John Henderson Lamb, M.D., *Sodium Bicarbonate: An Excellent Deodorant*, THE JOURNAL OF INVESTIGATIVE DERMATOLOGY 7(3) 131–33 (1946) (“Lamb”).

Ex. 1006, Bianchi et al., US 6,048,518 B2, issued April 11, 2000 (“Bianchi ’518”).

Ex. 1007, Wayback Machine archives, captured May 26, 2016, November 23, 2016, and November 1, 2015 respectively for: https:/www.nativecos.com, https:/www.nativecos.com/product/deodorant/, and https:/www.nativecos.com (“Native”).

Ex. 1008, Bianchi, US 2007/0166254 A1, published July 19, 2007 (“Bianchi ’254”).

Ex. 1009, Wayback Machine archive, captured January 26, 2017, for http:/soapdelinews.com/2016/11/easy-homemade-deodorantrecipe.html (“Easy Homemade”).

Petitioner relies on the declarations of Bozena B. Michniak-Kohn, Ph.D. (Exs. 1002 and 1086) to support its contentions.

Patent Owner relies on the declaration of Robert Lochhead, Ph.D. (Ex. 2011) to support its contentions.

F. Asserted Grounds of Unpatentability

Petitioner asserts that claims 1–15 would have been unpatentable on the following grounds:

Ground	Claim(s) Challenged	35 U.S.C. §	Reference(s)/Basis
I	1, 3–8, 11–14	102/103	Lesniak
II	2, 10	103	Lesniak, Phinney
III	9	103	Lesniak, Lamb
IV	15	103	Lesniak, Bianchi ’518,
V	1–2, 5–9, 11–15	103	Native, Bianchi ’254
VI	3, 4, 10	103	Native, Bianchi ’254, Easy Homemade

G. Claim Construction

The challenged claims should be read in light of the Specification, as it would be interpreted by one of ordinary skill in the art. *In re Suitco Surface, Inc.*, 603 F.3d 1255, 1260 (Fed. Cir. 2010). Thus, we generally give claim terms their ordinary and customary meaning. *See In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007) (“The ordinary and customary meaning is the meaning that the term would have to a person of ordinary skill in the art in question.” (internal quotation marks omitted)); *see also* 37 C.F.R. § 42.100(b) (stating that claims are construed in IPRs according to the same standard as used in federal court).

Petitioner does not argue for the express construction of any claim term. *See generally* Pet.

Patent Owner asserts that the preamble of independent claim 1 reciting “a deodorant stick” is limiting” as it provides antecedent basis for certain other claim elements, specifically, for the recitation of “said stick being free of an aluminum salt" and "said stick" having a particular hardness range. PO Resp. 8–10. We agree. Petitioner does not provide argument to the contrary. *See, e.g.*, Pet. 27 (arguing that Lesniak discloses a deodorant stick in the form of an antiperspirant stick).

We determine that no express construction of any other claim term is necessary to resolve the dispute between the parties. *Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) ("[W]e need only construe terms 'that are in controversy, and only to the extent necessary to resolve the controversy.'" (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999))). To the extent further discussion of the meaning any claim term is necessary to our decision, we provide that discussion below in our analysis of the asserted grounds of unpatentability.

H. Level of Ordinary Skill in the Art

Petitioner proposes that a person of ordinary skill in the art ("POSA" or "POSITA") at the time of the invention "would have (1) at least a B.S. or equivalent degree; and (2) at least two years of experience in cosmetics formulations." Pet. 27 (citing Ex. 1002 ¶¶ 26–28). Patent Owner does not dispute Petitioner's proposal about the POSA's qualifications. PO Resp. 7–8.

We adopt and apply Petitioner's proposed definition of a POSA, which does not appear to be inconsistent with the level of skill reflected in the asserted prior art.

III. ANALYSIS

A. Introduction

"In an *inter partes* review, the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable." *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify "with particularity . . . the evidence that supports the grounds for the challenge to each claim")). This burden of persuasion never shifts to the patent owner. *See Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015). Moreover, a petitioner should not "place the burden on [the Board] to sift through information presented by the Petitioners, determine where each element [of the challenged claims] is found in [the cited references], and identify any differences between the claimed subject matter and the teachings of [the cited references.]" *Google Inc. v. EveryMD.com LLC*, IPR2014-00347, Paper 9 at 25 (PTAB May 22, 2014).

Anticipation is a question of fact, as is the question of what a prior art reference teaches. *In re NTP, Inc.*, 654 F.3d 1279, 1297 (Fed. Cir. 2011). "Because the hallmark of anticipation is prior invention, the prior art reference—in order to anticipate under 35 U.S.C. § 102—must not only disclose all elements of the claim within the four corners of the document, but must also disclose those elements 'arranged as in the claim.'" *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359, 1369 (Fed. Cir. 2008) (quoting *Connell v. Sears, Roebuck & Co.*, 722 F.2d 1542, 1548 (Fed. Cir. 1983)). Whether a reference anticipates a claim is assessed from the skilled artisan's perspective. *See Dayco Prods., Inc. v. Total Containment, Inc.*, 329 F.3d 1358, 1368 (Fed. Cir. 2003) ("[T]he dispositive question regarding anticipation [i]s whether one skilled in the art would reasonably understand or infer from the [prior art reference's] teaching that every claim element was disclosed in that single reference." (quoting *In re Baxter Travenol Labs.*, 952 F.2d 388, 390 (Fed. Cir. 1991)))) (second and third alterations in original)).

A patent claim is unpatentable for obviousness if the differences between the claimed subject matter and the prior art are such that the subject matter, as a whole, would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). In *Graham v. John Deere Co. of Kan. City*, 383 U.S. 1 (1966), the Supreme Court set out a framework for assessing obviousness that requires consideration of four factors: (1) the "level of ordinary skill in the pertinent art," (2) the "scope and content of the prior art," (3) the "differences between the prior art and the claims at issue," and (4) "secondary considerations" of nonobviousness such as "commercial success, long felt but unsolved needs, failure of others, etc." *Id.* at 17–18; *KSR*, 550 U.S. at 407.

"Whether an ordinarily skilled artisan would have been motivated to modify the teachings of a reference is a question of fact." *WBIP, LLC v. Kohler Co.*, 829 F.3d 1317, 1327 (Fed. Cir. 2016). "[W]here a party argues a skilled artisan would have been motivated to combine references, it must show the artisan 'would have had a reasonable expectation of success from doing so.'" *Arctic Cat Inc. v. Bombardier Recreational Prods. Inc.*, 876 F.3d 1350, 1360–61 (Fed. Cir. 2017) (quoting *In re Cyclobenzaprine Hydrochloride Extended-Release Capsule Patent Litig.*, 676 F.3d 1063, 1068–69 (Fed. Cir. 2012)).

We analyze the asserted grounds of unpatentability in accordance with these principles to determine whether Petitioner has met its burden to establish a reasonable likelihood of success at trial.

B. Summary of the Cited Prior Art

1. Lesniak (Ex. 1003)

Lesniak is titled "NON-GREASY PERSONAL CARE COMPOSITIONS." Ex. 1003, code (54). Lesniak provides an anhydrous personal care composition, comprising

> (a) at least one ester, wherein said at least one ester is selected from one or more of the following groups: (i) triglycerides, (ii) diglycerides, (iii) monoglycerides, (iv) monoesters of dials, and (v) diesters of dials, and wherein a total concentration of said at least one ester is 45 to 60 weight %, based on the total weight of the composition; (b) a first wax having a melting point of 35° C. to 72° C. present in an amount of 5 to 25 weight %, based on the total weight of the composition; (c) a second wax having a melting point of 73° C. to 90° C. present in an amount of 0.5 to 15 weight% based on the total weight of the composition; (d) at least one plant oil having a saponification value of 150 to 275 mg KOH/g, wherein a total concentration of said at least one plant oil is from 5 to 35 weight%, based on the total weight of the composition.

Id. at 5:32–48. Lesniak describes that "[o]ptionally, the at least one ester is a combination of caprylic triglyceride and capric triglyceride" in ranges from 50–60 weight %, based on the total weight of the composition. *Id.* at 2:38–39, 3:42–44. "Optionally, the composition further comprises zinc oxide . . . present in a concentration of 10 to 30 weight %." *Id.* at 3:39–42. Lesniak describes that the composition may be in the form of a stick. *Id.* at 4:22. According to Lesniak, the compositions possess "good micro-robustness, . . . reduced greasiness and an improved after-feel (such as reduced tackiness and drag) on the skin as compared to other anhydrous compositions[, and] . . . are also physically stable at high and low temperatures." *Id.* at 5:50–54.

Lesniak discloses a desire to "to formulate such compositions from all-natural ingredients (i.e. '100% natural' compositions)." *Id.* at 5:29–31. Lesniak discloses that "the term 'natural compositions' indicates that the composition is formulated from ingredients which are extracted from the earth or sea, derived from plants, or synthesized by other organisms." *Id.* at 4:61–64. Lesniak discloses that the composition may comprise "essential oils which may be used as natural fragrances." *Id.* at 8:38.

2. Phinney (Ex. 1004)

Phinney is titled "DEODORANT FORMULATION." Ex. 1004, code (54). Phinney relates to a deodorant "formulation and preparation thereof with natural ingredients in the absence of heavy metal complexes and/or petroleum based or related compounds." *Id.* at 1:7–9. Phinney discloses finding that "a mixture of the hydroxides of zinc and magnesium could provide long lasting protection against odour due to perspiration . . . and are dispersible in aerosols, moisturizing creams, sticks, sprays and roll-on type applicators." *Id.* at 1:17–22.

Phinney explains that zinc and aluminum oxide are "very unreactive against perspiration odour due to insolubility and low hydrolysis rate." *Id.* at 2:35–36. According to Phinney, "magnesium hydroxide, . . . has low solubility and therefore does not 'wash away' under heavy perspiration" and thus "provides 16–20 hour protection and more for a broad cross section of people." *Id.* at 5:47–51. Phinney discloses an embodiment of a specific formulation prepared as a solid stick that "includes magnesium oxide in an amount of between 1% and 2.5% by weight and zinc oxide in an amount of between 0.5% and 1.5% by weight." *Id.* at 6:32–34, 6:11, 8:46–57 (claim 1).

3. Lamb (Ex. 1005)

Lamb is an article titled "SODIUM BICARBONATE: AN EXCELLENT DEODORANT" that was published in *The Journal of Investigative Dermatology*. Ex. 1005, 131. Lamb describes a study that evaluated the effectiveness of baking soda as an underarm deodorant in 90 patients. Ex. 1005, 131–33. Lamb hypothesizes that the perspiration's free fatty acids, which possess a disagreeable odor, forms a less malodorous sodium salt in the presence of sodium bicarbonate. *Id.* at 133. Based on the results of the study, Lamb concludes that "bicarbonate of soda is a valuable underarm deodorant for common daily usage." *Id.* Lamb observed that "[n]o furunculosis," or boils, "developed in over 90 cases which have used bicarbonate of soda over periods of from four weeks to a year's duration." *Id.*

4. Bianchi '518 (Ex. 1006)

Bianchi '518 is titled "LOW RESIDUE SOLID ANTIPERSPIRANT." Ex. 1006, code (54). Bianchi '518 "relates to antiperspirant sticks, substantially free of water and long chain fatty alcohols, which provide the user with excellent antiperspirant efficacy, reduced residue when the composition is first applied to the skin, reduced residue after dry down, high temperature stability, and excellent cosmetics and aesthetics." *Id.* at 1:8–13.

Bianchi '518 discloses that the antiperspirant composition possesses a specification of "Penetration-ASTM D-1321@ 100/77/5 10–16." *Id.* at 2:57–58. Bianchi '518 explains that

> [o]ne important physical parameter affecting consumer acceptance of suspensoid antiperspirant sticks of the invention is hardness as measured by needle penetration. Typical compositions of the invention will range in hardness from about 7.0 to about 14.0 mm as shown by the above mentioned needle penetration test. Products outside this range will typically generate low consumer acceptance ratings.

Id. at 6:23–29.

5. Native (Ex. 1007)

Native is a printed publication from a website advertising a deodorant product sold in 2016. Ex. 1007, 5–8, 11–12, 14–18. Native is advertised as a "Deodorant That Isn't A Chemistry Experiment," which contains "NATURAL INGREDIENTS" and is "ALUMINUM-FREE." *Id.* at 5. Native lists each ingredient for the Native deodorant stick product, including, *inter alia*, shea butter, beeswax, baking soda, and stearyl alcohol. *Id.* at 6, 11.

The picture on page 5 of Native, reproduced below, illustrates a deodorant product. *Id.* at 5.

The picture of the Native deodorant product, reproduced above, illustrates the deodorant stick in a supine position with the cap removed. *Id.*

6. Bianchi ’254 (Ex. 1008)

Bianchi ’254 is titled “ANTIPERSPIRANT STICK COMPOSITIONS.” Ex. 1008, code (54). Bianchi ’254 discloses “antiperspirant stick compositions comprising specified amounts of antiperspirant active; carrier oil comprising volatile silicone oil and non-volatile masking oil; structurant comprising fatty alcohol and cosmetically acceptable wax having a melting point of 75 to 95° C., a portion of wax comprises polyethylene in specified amounts.” *Id.* at code (57). Bianchi ’254 addresses

> a need for non-whitening fatty alcohol-containing antiperspirant sticks, particularly sticks containing fatty alcohol at levels of 12% by weight or above, more particularly, 15 wt % or above. Also desired are non-whitening fatty alcohol-containing sticks that have desirable antiperspirant payouts, while providing a ‘clean’, non-sticky feel. There also remains a need for a non-whitening fatty-alcohol containing stick that provides desirable payout as well as being thermally stable and resistant to dome collapse at temperatures of 50° C.

Id. ¶ 10. Table 5 of Bianchi ’254 provides eight different formulations of antiperspirant sticks with each of their respective components and amounts listed. *Id.* ¶ 131.

Bianchi ’254 explains that the “[t]he hardness of the sticks can be measured in a needle penetration test.” *Id.* ¶ 26. Bianchi ’254 describes the method for measuring stick hardness using a “brass/steel tapered needle (DIN 51-579)” depth on six locations of the stick, “with the hardness values being reported as an average of the six measurements.” *Id.* ¶ 117.

7. Easy Homemade (Ex. 1009)

Easy Homemade is a printed publication of a blog post titled "Easy Homemade Deodorant Recipe with just 3-Ingredients." Ex. 1009, 5. Easy Homemade describes the blogger's first experience with natural deodorant, but falling "victim to the burning fiery rash you get after using deodorant with baking soda." *Id.* at 6. Easy Homemade describes that the blogger discovered that she "could swap the baking soda with magnesium hydroxide - basically the same thing as Milk of Magnesia - and it worked just as well." *Id.* The Easy Homemade blogger describes melting "two natural deodorants" that are aluminum-free, and making a "[h]uge difference" in each deodorants' performance by adding "arrowroot powder and magnesium hydroxide to them." *Id.* at 7. Easy Homemade provides a deodorant recipe and lists "2.15 oz. natural aluminium free deodorant (base) of choice[,] 1 oz. arrowroot powder[,] .5 oz. magnesium hydroxide[,]" to which one may "[a]dd essential oils if desired." *Id.* at 7–8.

C. Ground I: Anticipated or Obvious over Lesniak

Petitioner asserts that claims 1, 3–8, and 11–14 are unpatentable under 35 U.S.C. § 102 as anticipated or, alternatively, under 35 U.S.C. § 103 as obvious over Lesniak. Pet. 27–34. In support of its assertion, Petitioner provides a detailed discussion and claim chart explaining how each claim limitation is disclosed by Lesniak. *Id.* Petitioner's contentions are supported by the declaration of Dr. Michniak-Kohn. *See, e.g.*, Ex. 1002 ¶¶ 57–96.

Patent Owner disagrees and provides several arguments Lesniak does anticipate or render obvious the challenged claims. PO Resp. 14–47; Sur-reply 17–22.

After considering all of the arguments and cited evidence of record, we determine that preponderance of the evidence supports the conclusion that Lesniak renders the challenged claims unpatentable. We address the parties' arguments below. We begin our analysis addressing the disputed elements of claim 1.

1. Independent claim 1

Claim 1 recites a deodorant stick comprising at least about 25% of a liquid triglyceride, an antimicrobial, and a primary structurant with a melting point of at least about 50° C. Claim 1 further requires that the deodorant stick be free of an aluminum salt.

Petitioner contends that Lesniak discloses each element of claim 1. Pet. 27–28. In particular, Petitioner contends that Lesniak discloses a deodorant stick in the form of an antiperspirant stick that does not contain aluminum salts. *Id.* (citing Ex. 1003, 1:16–35, 4:22, 5:5–32, 10:27, 10:32–33). Petitioner further contends that Lesniak discloses that its compositions may comprise at least about 35 wt% of a liquid triglyceride (i.e. caprylic/capric triglyceride, citing Ex. 1003, 2:38–39, 9:45–46); about 8 to 19 wt% of a primary structurant with a melting point of at least about 50° C (i.e. beeswax, citing *id.* at 1:39–54, 2:5–7, 2:24–25); and at least one antimicrobial (i.e. zinc oxide, citing *id.* at 3:39–4:21, 8:31–9:59). *Id.* at 28.

Patent Owner responds to Petitioner's challenge with several arguments, which we address below. PO Resp. 16–53.

a) Whether Lesniak discloses a deodorant stick

Patent Owner first contends that "Lesniak does not disclose a 'deodorant stick,' as required by the preamble of the sole independent claim." PO Resp. 15. Patent Owner acknowledges that Lesniak suggests that "[t]he composition can also be an antiperspirant" (Ex. 1003, 10:27–29), but contends that Lesniak fails to "specify any form of antiperspirant, which can come in a variety of forms, including aerosol sprays, creams, liquids, and gels." PO Resp. 19 (citing Ex. 2011 ¶ 55). Patent Owner further contends that Lesniak's "passing references to 'antiperspirant' fail to convert Lesniak into something other than a reference about the embodied sunscreens and diaper creams." *Id.*

We are not persuaded by Patent Owner's arguments. Rather, we agree with Petitioner that Lesniak teaches anhydrous personal care compositions where "[t]he composition can . . . be an antiperspirant" and "can also be a stick." Ex. 1003, 9:45, 10:27. Thus, Lesniak discloses a deodorant in the form of an antiperspirant. Reply 3 (citing Ex. 1036, 2 ("all antiperspirants are deodorants"); Ex. 1037, 246 (same); Ex. 1084, 33 ("the main mechanisms through which deodorant products intended to combat body odors . . . were the usage of antiperspirants that prevent perspiration"). In light of those express disclosures, we are unpersuaded by Patent Owner's arguments that Lesniak fails to disclose a "deodorant stick."

Additionally, Patent Owner contends that Lesniak's compositions contain large amounts of zinc oxide (up to 20%) and that a POSA would have understood that "zinc oxide, particularly in the high amounts present in Lesniak, is not commonly used or useful in deodorants." PO Resp. 18–19 (citing Ex. 2011 ¶ 44). We are not persuaded. The evidence of record does not support a conclusion that compositions comprising 10 to 30% zinc oxide such as disclosed in Lesniak would not be appropriate for deodorants. Again, we credit the testimony of Dr. Michniak-Kohn "that deodorants were known to contain ranges of zinc oxide that encompass the amount of zinc oxide in Lesniak's compositions." Ex. 1086 ¶ 9 (citing Ex. 1001, 7:7–19; Ex. 1072, 1–2 (disclosing deodorants having "from 0.5 to 20%" zinc oxide)).

b) Whether antiperspirants must contain aluminum

Patent Owner contends that Lesniak's antiperspirants must contain aluminum. PO Resp. 19–21. Specifically, Patent Owner contends that,

> To a POSA, the term "antiperspirant" means a composition containing an aluminum salt or complex—the only antiperspirant actives approved by the FDA. *See* 37 C.F.R. 350. As Dr. Lochhead explains, a POSA would have understood that the U.S. Food and Drug Administration regulates antiperspirants as drugs and regards deodorants as cosmetics. (Ex. 2011 at ¶ 48.) A POSA would have further understood Lesniak's disclosure of "antiperspirant" as requiring the use of an antiperspirant active, and the only antiperspirant actives approved by the FDA all contain aluminum. (*Id*. at ¶ 48.)

PO Resp. 20–21 (footnote omitted); *see also* Sur-reply 18–19 ("a POSA's understanding of 'antiperspirants,' now and at the time of Lesniak, is that they contain aluminum" (citing Ex. 1081,[1] 60:12–14 ("[A] POSA would understand if you're talking about an antiperspirant, it has to have aluminum."); Ex. 2011 ¶¶ 48–51)).

We are not persuaded by Patent Owner's argument. Rather, we agree with Petitioner that, "[w]hile [Patent Owner] emphasizes that the FDA requires antiperspirants marketed in the U.S. to contain aluminum, anticipation and obviousness are distinct from FDA regulatory requirements." Reply 4 (citing *Persion Pharms. LLC v. Alvogen Malta Ops. Ltd*., 945 F.3d 1184, 1193 (Fed. Cir. 2019) ("FDA's approval requirements do not undermine the force of the evidence as to obviousness"). Indeed, the evidence of record shows that zinc compounds were known antiperspirant

[1] Ex. 1081, Dr. Robert Lochhead Deposition Transcript, July 10, 2025.

actives that could be alternatives to aluminum. *Id.* at 5 (citing Ex. 1003, 3:39–4:21, 8:46–48, 10:27 (describing zinc-based antiperspirant compositions); Ex. 1014 ¶ 20 ("[A]ntiperspirant actives include astringent metallic salts, particularly inorganic and organic salts of aluminum, zirconium, and zinc, as well as mixtures thereof.")).

c) Whether Lesniak enables a deodorant stick

Patent Owner contends that Lesniak fails to provide sufficient guidance as to how to formulate a stable deodorant or antiperspirant stick, and therefore, is non-enabling. *Id.* at 24–31 (citing Ex. 2011 ¶¶ 52–66). In particular, Patent Owner contends that "the only 'stick' formulation Lesniak expressly teaches is a sunscreen stick." PO Resp. 25 (citing Ex. 1003, Table 7). Patent Owner also directs our attention to the results disclosed in Lesniak's Table 2 and argues that Lesniak's compositions suffered from stability issues. *Id.* at 26–30; *see also* Sur-reply 17 ("Lesniak's Table 2 shows that even small changes in a composition can yield unstable results," demonstrating that the art is unpredictable.). Patent Owner concludes that "Lesniak provides no teaching of a stable 'deodorant stick,' no teaching of a stable 'antiperspirant stick,' and at least nine different sunscreen formulas with slight variations that are all unstable." PO Resp. 30 (citing Ex. 2011 ¶¶ 52–66).

We find that the evidence of record fails to support Patent Owner's argument that the prior art fails to enable a stable deodorant stick. Rather, we are persuaded by Petitioner that Lesniak sufficiently teaches compositions with the relevant ingredients and achieves stable compositions through the use of a second wax having a melting point of 73 to 90° C. Reply 7–8 (citing Ex. 1003, 1:39–54). For example, Lesniak's compositions (nos. 8 and 11) that used such a second wax, namely CastorWax MP80, are disclosed as sufficiently stable. Ex. 1003, Table 2. Here, we further credit the testimony of Dr. Michniak-Kohn that

> Lesniak teaches how to make stable compositions by including a secondary wax, such as castor wax, with a sufficiently high melting point. Ex. 1003, 1:39-54, 5:23-32, Table 2. Both of the compositions (nos. 8 and 11) tested by Lesniak that had such a wax passed the stability tests.

Ex. 1086 ¶ 7; *see also* Reply 7–8 ("Lesniak overcame any stability issues, as it expressly discloses compositions that are 'physically stable at high and low temperatures.'") (citing Ex. 1003, 5:23–32)).

In view of the above, we determine that Lesniak provides sufficient disclosure to enable a person of ordinary skill in the art to formulate deodorant stick within the parameters of claim 1.

d) Conclusion as to claim 1

Having considered the parties' positions and evidence of record, summarized above, we are persuaded by Petitioner's analysis that Lesniak discloses all elements of claim 1. We are not persuaded to Patent Owner's arguments to the contrary. Accordingly, we determine that Lesniak anticipates claim 1.

D. Ground II: Obviousness over Lesniak and Phinney

Petitioner asserts that claims 2 and 10 are unpatentable under 35 U.S.C. § 103 as obvious over the combination of Lesniak and Phinney. Claims 2 and 10 depend from claim 1 and further require that the antimicrobial to be selected from, among other things, magnesium carbonate and magnesium hydroxide. Ex. 1001, 15:44–46, 16:1–14, 16:29–30. Petitioner concedes that Lesniak does not expressly teach the use of magnesium hydroxide. Pet. 34. However, Petitioner asserts that, in view of

Phinney, a POSA would have found it obvious to add magnesium hydroxide to the composition of Lesniak. *Id.* Phinney states that metal oxides, such as zinc oxide, are not particularly efficacious antimicrobials in deodorants when used alone and describes using magnesium hydroxide as an antimicrobial in a deodorant stick in combination with zinc oxide. *Id.* at 35.

Patent Owner contends that Lesniak and Phinney describe entirely different uses for their metal oxides/hydroxides. PO Resp. 33. In particular, Patent Owner contends that Lesniak includes zinc oxide in sunscreen and diaper cream compositions as a protective barrier, not for odor control. *Id.* Moreover, Patent Owner contends that "Lesniak's formulas with high concentrations of zinc oxide [are] unsuitable for use as an underarm deodorant or antiperspirant." *Id.* (citing Ex. 2011 ¶ 71; Ex. 1004, 2:33–36 ("[Z]inc oxide . . . [is] very unreactive against perspiration odour."), 3:20–21). Thus, a "POSA would not have been motivated to look at Phinney to modify Lesniak or incorporate any additional antimicrobial into Lesniak 'to improve odor protection,' as odor protection was not a goal of Lesniak's zinc oxide in the first place." *Id.*

Furthermore, Patent Owner contends that Phinney does not suggest a combination of zinc oxide and magnesium hydroxide. *Id.* at 34–35. Rather, according to Patent Owner,

> Petitioner cherry picks the single ingredient magnesium hydroxide from Phinney's disclosure of a three- or four-ingredient combination to combat odor. Phinney explains that to combat odor, its compositions contain a combination of (1) magnesium hydroxide, and (2) zinc hydroxide, and (3) potassium chloride and/or potassium sulfate. (Ex. 1004 at 8:7-31, cl. 1.) There are two significant aspects to this multi-part odor fighting combination: the requirement of zinc hydroxide, not zinc oxide; and the requirement of potassium chloride/sulfate. (Ex. 2011 at ¶ 72.)

PO Resp. 34 (emphasis omitted). Patent Owner further contends that Lesniak and Phinney are incompatible, arguing that Lesniak's compositions are anhydrous while Phinney's compositions requires water. *Id.* at 35 (citing 2011 ¶ 74); *see also* Ex. 1004, 4:48–51, 8:22–41 (explaining magnesium oxide and zinc oxide are both mixed with water to become their respective hydroxides, and that they "must be completely converted to their hydroxides or skin irritation will occur"); Sur-reply 20–21.

We have considered Patent Owner's arguments but find, on this record, Petitioner to have the better position. In particular, we are persuaded that a POSA would have been motivated to add magnesium hydroxide to Lesniak's compositions because

> Phinney teaches that: (i) zinc oxide has "low activity" as a "deodorant agent"; (ii) magnesium hydroxide is "an effective deodorant"; and (iii) zinc and magnesium compounds have synergistic effects, hence motivating a POSA to add magnesium hydroxide to Lesniak's zinc oxide deodorants. Ex. 1004, 1:13-3:16, 5:61-67.

Reply 10; *see also* PO Resp. 34–35 (arguing that a POSA would have found it obvious to add magnesium hydroxide to the composition of Lesniak because zinc oxide is not a particularly efficacious antimicrobial in deodorants when used alone, whereas magnesium hydroxide is an effective deodorant).

We acknowledge that Phinney suggests mixing zinc oxide with water in order to convert zinc oxide to zinc hydroxide. *See*, *e.g.*, Ex. 1004, 1:17–19 ("It was also found that a mixture of the hydroxides of zinc and magnesium could provide long lasting protection against odour due to perspiration."). The evidence of record, however, shows that both magnesium hydroxide and zinc oxide were known options for deodorant antimicrobials, and claims 2 and 10 do not require a minimal level of effectiveness such that the combination of magnesium hydroxide and zinc oxide would need to outperform any other combination including the mixture of zinc and magnesium hydroxides. *See In re Fulton*, 391 F.3d 1195, 1200 (Fed. Cir. 2004) ("[O]ur case law does not require that a particular combination must be the preferred, or the most desirable, combination[.]"). Here, we also credit the testimony of Dr. Michniak-Kohn that "POSA would understand that the water-based vs. anhydrous distinction is not relevant, as a POSA would understand that magnesium hydroxide would still function as an antimicrobial in anhydrous compositions such as those disclosed by Lesniak." Ex. 1086 ¶ 9 (citing Ex. 1072, 1–2, 5 (disclosing deodorants without added water and with zinc oxide and/or magnesium hydroxide)). Accordingly, we are persuaded by Petitioner that a POSA would have been motivated to incorporate magnesium hydroxide into Lesniak's composition following Phinney's guidance, as this combination would improve odor protection while maintaining the 'natural' composition intended by Lesniak. PO Resp. 35–36 (citing Ex. 1002 ¶¶ 83–84).

Ultimately, however, it does not matter whether Phinney teaches a composition that contains water. "[I]t is not necessary that [two pieces of prior art] be physically combinable to render obvious" the claimed subject matter. *Allied Erecting and Dismantling Co., Inc. v. Genesis Attachments, LLC*, 825 F.3d 1373, 1381 (Fed. Cir. 2016) (quoting *In re Sneed*, 710 F.2d 1544, 1550 (Fed. Cir. 1983)). Rather, the question is "whether the claimed inventions are rendered obvious by the teachings of the prior art as a whole." *In re Etter*, 756 F.2d 852, 859 (Fed. Cir. 1985) (noting that whether one prior art reference can be incorporated into another is "basically irrelevant.") Lesniak and Phinney undisputedly teach that magnesium hydroxide and zinc oxide were known ingredients in personal care products including deodorant sticks. The inclusion of magnesium hydroxide and zinc oxide into a single product therefore represents, absent evidence to the contrary, "a combination of familiar elements according to known methods that does no more than yield predictable results." *Agrizap, Inc. v. Woodstream Corp.*, 520 F.3d 1337, 1344 (Fed. Cir. 2008).

For the reasons discussed above, we conclude that Petitioner has proved by a preponderance of evidence that the combined teachings of Lesniak and Phinney render obvious claims 2 and 10.

E. Ground III: Obviousness over Lesniak and Lamb

Petitioner asserts that claim 9 is unpatentable under 35 U.S.C. § 103 as obvious over the combination of Lesniak and Lamb. Claim 9 depends from claim 1 and further requires that the antimicrobial comprises baking soda. Ex. 1001, 16:28–29. Petitioner asserts that Lamb discloses a deodorant stick that comprises baking soda as an antimicrobial and teaches that baking soda is an "excellent deodorant" and "a valuable underarm deodorant for common daily usage." Pet. 38 (citing Ex. 1005, 132–133).

Petitioner further provides a reason to combine Lesniak and Lamb. *Id.* at 39–40. Specifically, Petitioner contends that,

> While Lesniak discloses natural deodorant compositions comprising zinc oxide (a known antimicrobial), it does not disclose the use of baking soda as an antimicrobial. By the time of the filing of the '915 Patent, a POSA would have known that zinc oxide was not a strong antimicrobial when used alone in deodorant formulations. Also, a POSA would have understood that deodorant compositions often use multiple antimicrobials. A POSA would have found it obvious to use additional or alternative antimicrobials in Lesniak's deodorant compositions to enhance odor protection.

Id. at 39.

Patent Owner contends that POSA would not have been motivated to add baking soda to Lesniak's compositions because baking soda was known to cause irritation for some users. PO Resp. 45–47.

We are not persuaded by Patent Owner's argument. Lamb expressly teaches baking soda as an effective ingredient in a deodorant. Ex. 1005, 131–33. Baking soda is also known to cause irritation that may motivate some to remove it from deodorants. *See* Ex. 2011 ¶¶ 97–102. Thus, the specific set of advantages and disadvantages for using baking soda in a deodorant was known to a person of skill in the art. On this record, the inclusion or exclusion of baking soda presents no unexpected result and the evidence of record establishes that it would have been an obvious matter of design choice within the skill of the art to balance the advantages and disadvantages associated therewith. *See Medichem, S. A. v. Rolabo, S.L.*, 437 F.3d 1157, 1165 (Fed. Cir. 2006) ("a given course of action often has simultaneous advantages and disadvantages, and this does not necessarily obviate motivation to combine"); *see also Winner Int'l Royalty Corp. v. Wang*, 202 F.3d 1340, 1349 n.8 (Fed. Cir. 2000) ("The fact that the motivating benefit comes at the expense of another benefit, however, should not nullify its use as a basis to modify the disclosure of one reference with the teachings of another.")

In view of the above, we determine that the preponderance of the evidence establishes that claim 9 would have been obvious over the combination of Lesniak, Phinney, and Lamb.

F. Ground IV: Obviousness over Lesniak and Bianchi ’518

Petitioner asserts that claim 15 is unpatentable under 35 U.S.C. § 103 as obvious over the combination of Lesniak and Bianchi ’518. Claim 15 depends from claim 1 and further requires the deodorant stick to have a hardness from about 80 mm*10 to about 140 mm*10, as measured by penetration with ASTM-D 1321 needle. Ex. 1001, 16:43–45. Petitioner acknowledges that Lesniak does not expressly teach hardness as measured by penetration with an ASTM-D 1321 needle. Pet. 41. For this element of claim 15, Petitioner relies on Bianchi ’518, which teaches that consumers prefer products falling within a hardness range of about 7.0 mm to about 14.0 mm (equivalent to about 70 mm*10 to about 140 mm*10), as measured by penetration with a standard ASTM-D 1321 needle. *Id.* at 41–42 (citing Ex. 1006, 2:56–58, 6:23–29). Petitioner contends that it would be obvious to formulate Lesniak’s deodorant stick to be within the claimed hardness range, which overlaps with ranges known to be acceptable to consumers. *Id.* at 42–44 (citing Ex. 1002 ¶¶ 97–100). Furthermore, “[a] POSA would have understood that the hardness could be adjusted by altering the amount of the primary structurant and/or by incorporating secondary structurants. *Id.* at 44 (citing Ex. 1002 ¶ 101).

Patent Owner contends that the standard hardness test referenced in Bianchi-518 is different from the test described in the ’915 patent and contends that

> the Bianchi hardness test uses a needle and shaft with *double the weight* of the needle and shaft used in the ’915 Patent (100 grams vs. 50 grams), and thus its so-called same or similar hardness ranges reflect that the Bianchi compositions are actually *significantly harder* than those claimed in the ’915 Patent (i.e., it requires twice the weight of the penetration needle assembly to yield similar penetration depths).

PO Resp. 40 (citing Ex. 2011 ¶¶ 83, 85; Ex. 2018,[2] 5–6 ("Place a 50-g weight above the penetrometer needle, making a total load of 100 ± 0.015 g for the needle and all attachments.")). Patent Owner supports its argument with the testimony of Dr. Lochhead, who explains that

> The '915 Patent uses a different testing method. The '915 Patent explains that its invention exhibits a hardness in the range of "80 mm*10 to about 140 mm*10," as measured using an ASTM D-1321 penetration needle in a test similar to ASTM D-1321, but with one significant difference: "the total weight of the shaft and needle is 50±0.2 grams when the shaft is in free fall." ('915 at 12:24-25.) The '915 Patent's method does not include the "50-g weight" added to the needle assembly as an additional load, which is used in the standard ASTM D-1321 test utilized by Bianchi-518.

Ex. 2011 ¶ 86. Patent Owner further contends that the evidence of record fails to support a conclusion that Bianchi-518's test and the '915 patent's test would produce overlapping hardness ranges.[3] PO Resp. 41.

We find that the evidence of record fails to support Patent Owner's position that the '915 patent discloses a unique hardness test. As noted by Petitioner, the '915 patent discloses that hardness was measured by "a needle penetration test under ASTM D-1321 (as described herein)." Ex. 1001, 3:8–9. The '915 patent describes that its hardness test was based on ASTM D-1321 and used penetration needles "as specified for the ASTM D 1321 D 1321/DIN 51 579, Officially certified, Taper-Tipped needle, No.

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[2] Ex. 2018, ASTM D-1321 Needle Penetration Test Standard (1996).

[3] Patent Owner also asserts that Petitioner improperly submitted evidence with its Reply to correlate the hardness test methods of the Bianchi references and the challenged claims. Because we do not agree on the record before us that these test methods are different, we do not rely on Petitioner's newly submitted evidence and will not further address Patent Owner's assertions. Sur-reply 8–14.

H-1310, Humboldt Mfg." *Id.* at 11:44–12:15. The '915 patent further discloses that "[p]eriodic calibration should confirm: . . . the total weight of the shaft and needle is 50±0.2 grams when the shaft is in free fall." *Id.* at 12:20–26. That disclosure for periodic calibration coincides with the requirement under the ASTM D-1321 test that "[t]he final weight of the needle shall be 2.5 ± 0.05 g. The total weight of the plunger shall be 47.5 ± 0.05 g." Ex. 2018, 5. However, the ASTM D-1321 test also makes clear that a 50-gram weight is placed "above the penetrometer needle, making a total load of 100 ± 0.15 g for the needle and all attachments," which is consistent with Bianchi-518's disclosure of a test using 100 grams of weight. Ex. 1006, 2:58, 6:25–28. The '915 patent, on the other hand, fails to expressly disclose whether the test was performed without the required 50-gram weight placed above the penetrometer needle, and its silence to that fact is insufficient to persuade us that a different test was ultimately intended or performed. Rather, we are persuaded by the following testimony of Dr. Michniak-Kohn:

> Dr. Lochhead states that he believes the claimed recitation of a hardness range "as measured by penetration with ASTM D-1321 needle" should not be understood to indicate the ASTM D-1321 standard, but a modification of that standard. Ex. 2011, 41. As a basis for his opinion. Dr. Lochhead focuses on language in the '915 Patent that "the total weight of the shaft and needle is 50±0.2 grams." Ex. 2011, ¶25. But he omits the preceding language, which states that this is the weight to be used for "[p]eriodic calibration" to "confirm" that the equipment is in proper working order. Ex. 1001, 12:20-29. That "[p]eriodic calibration" section precedes a section identifying what needs to be confirmed "[a]t the time of use." *Id.*, 12:30-37. Those two sections would suggest to a POSA routine checking of equipment. Only after those two sections does the specification set forth the test. *Id.*, 12:38-67 ("Sample Preparation and Measurement"). Moreover, the "[p]eriodic calibration" comports

> with the ASTM D-1321 standard, which likewise requires that the "weight of the needle shall be 2.5 g" and the "weight of the plunger shall be 47.5 g," for a 50-g total weight of the needle assembly. Ex. 2018, 5. Although the ASTM D-1321 standard further describes using an additional 50-g weight during the test, a POSA would not read the '915 Patent to exclude that weight. Rather, a POSA would consult the details of the ASTM D-1321 standard identified by the '915 Patent to ensure compliance.

Ex. 1086 ¶ 3 (alterations in original).

Accordingly, having considered the parties' positions and evidence of record, summarized above, we determine that a POSA would have been motivated to formulate Lesniak's deodorant stick to be within the claimed hardness range, which overlaps with ranges known to be acceptable to consumers as disclosed in Bianchi '518. Pet. 41–44 (citing Ex. 1002 ¶¶ 97–101).

G. Ground V: Obviousness over Native and Bianchi '254

Petitioner asserts that claims 1–2, 5–9, and 11–15 are obvious over the combination of Native and Bianchi '254. Pet. 45–63. In support of this assertion, Petitioner relies on Native for its teaching of "a natural, aluminum-free deodorant stick" marketed and sold in 2016. *Id.* at 45 (citing Ex. 1007, 5, 11). Petitioner contends that a POSA "looking to formulate new deodorant compositions would have looked to existing formulations and would have understood that commercial deodorant brands typically provided ingredient lists of their products on their websites." *Id.* at 43. Additionally, Petitioner provides a detailed discussion and claim chart explaining how each claim limitation is disclosed by the combination of Native and Bianchi '254. *Id.* at 49–63. Petitioner's contentions are further supported by the declaration of Dr. Michniak-Kohn. *See* Ex. 1002 ¶¶ 110–164.

Patent Owner disagrees and provides several arguments for why the combination of Native and Bianchi ’254 fails. PO Resp. 53–71; Sur-reply 23–25.

After considering the parties’ arguments and cited evidence of record, we determine that the preponderance of the evidence supports the conclusion that the combination of Native and Bianchi ’254 renders obvious claims 1–2, 5–9, and 11–15. We address the parties’ arguments below. We begin our analysis addressing the disputed elements of claim 1.

1. Independent Claim 1

Petitioner contends that Native discloses the features of deodorant stick compositions according to claim 1. For example, as the Petition discusses, Native discloses an aluminum-free deodorant stick including an antimicrobial, a primary structurants with a melting point of at least 50° C, and a liquid triglyceride. Pet. 49–54. Petitioner concedes that Native does not expressly teach the percentage of each ingredient used such as the percentage of liquid triglyceride (carrier oil). Pet. 46. For those elements of the dependent claims, Petitioner contends that

> Native sets forth the ingredients in a natural, aluminum-free deodorant stick that was being marketed and sold in at least 2016. Ex. 1007, 5, 11. The deodorant stick composition is comprised of caprylic/capric triglyceride, baking soda, stearyl alcohol, hydrogenated castor oil, polyglycerol-3 beeswax, and jojoba esters. *Id.* Silicones, aluminum salts, and synthetic fragrances are not included in the composition. *Id.* As a POSA would have understood, Native lists its deodorant stick ingredients in order from greatest to least concentration (Ex. 1002, ¶107 (citing Ex. 1046, 164); however, it does not expressly teach the percentage of each ingredient used, including the percentage of liquid triglyceride (i.e., carrier oil), stearyl alcohol (i.e., the primary structurant), or hydrogenated castor oil (castor wax) (i.e., secondary structurants). Ex. 1007, 11. . . . In the absence of

> express teachings on these points, a POSA would have looked to other publications in the field containing examples of weight percentages for those known components of deodorant sticks.

Pet. 46–47; *see also id.* at 49–54 (claim chart).

Petitioner contends also that "[a] POSA looking to formulate new deodorant compositions would have looked to existing formulations and would have understood that commercial deodorant brands typically provided ingredient lists of their products on their websites." *Id.* at 45. For example, Petitioner relies on Bianchi '254 to teach the appropriate concentrations of the ingredients of the deodorant stick such as the percentages of liquid triglyceride, a carrier oil. *See id.* at 55–56. Petitioner further contends that

> a POSA would have been motivated to use at least about 25% of a liquid triglyceride in a deodorant stick, and would have had a reasonable expectation of success in doing so, inasmuch as (i) Bianchi '254 already suggests the same with its 20-40% amount, (ii) a POSA would have, based on Bianchi '254's overall oil teachings, used even more of the liquid triglyceride when making a stick with only that oil carrier (as taught by Native).
>
> The overlap between the claimed liquid triglyceride range and the ranges in the prior art renders the claim presumptively obvious. The '915 Patent discloses nothing critical or unexpected regarding the claimed range relative to the prior art ranges. Ex. 1002, ¶¶114-118.

Pet. 52.

Patent Owner contends that "[a] POSA seeking to develop an aluminum-free deodorant stick would not have been motivated to implement the weight percentages disclosed in Bianchi-254, which describes and claims antiperspirant formulations containing about 50% by weight of aluminum and silicone." PO Resp. 50 (citing Ex. 2011 ¶ 108.) Patent Owner further contends that Petitioner "does not explain what would have motivated a POSA to have implemented the weight-percentages from a non-natural antiperspirant stick into the natural, aluminum-free stick of Native (if it is even possible), or what a POSA would have substituted for aluminum and volatile silicone in order to preserve the same weight percentages without the presence of the aluminum active and silicones." *Id.* 50–51 (citing Ex. 2011 ¶ 109–110.)

Having considered the parties' positions and evidence of record, summarized above, we find Petitioner to have the better position. First, we note that Petitioner looks to the teachings of Bianchi '254 to inform the appropriate amount for the ingredients described in Native, not to alter Native's formulation. *See* Pet. 46 (stating "it would be obvious to formulate a deodorant stick such as the stick disclosed by Native using ingredient concentrations as taught by Bianchi '254"). Therefore, Patent Owner's argument that the praise for the Native product would not incentivize a POSA to change the Native formulation misses the mark. Petitioner does not seek to change Native's formulation, but verify the amount of the listed ingredients. *Id.* In any event, as Petitioner notes, a reason to arrive at a claimed invention does not have to come from a particular identification of deficiencies in Native itself. As the Supreme Court noted in *KSR*, motivation may come from "any need or problem known in the field of endeavor at the time of invention and addressed by the patent." 550 U.S. at 420; *see* Reply 20.

Furthermore, as Petitioner notes, both Native and Bianchi '254 are in the same field as the '915 patent—personal care compositions, including deodorants and/or antiperspirants. Pet. 49. There is nothing in the record that we have found, nor has Patent Owner pointed us to any such evidence, to distinguish natural, aluminum-free deodorants from antiperspirants containing aluminum in such a way that ingredients of one would not be used for the same purpose in the other. As the Supreme Court noted in *KSR*, "[t]he combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results." 550 U.S. at 416. In fact, as Petitioner points out, Patent Owner in the '915 patent "incorporates references disclosing aluminum-containing antiperspirants for their disclosure of 'suitable' structurants, emulsifiers, antimicrobials, and other ingredients." Reply 23 (citing Ex. 1001, 5:66–6:4, 10:57–64; Ex. 1041; Ex. 1083; Ex. 1081, 46:22–52:22).

Moreover, Dr. Michniak-Kohn persuasively asserts on the record before us that when looking at particular ingredients of a formulation, a POSA would look to other references in the same field—personal care compositions, including deodorants and/or antiperspirants—to determine how much of each ingredient should be used. Ex. 1002 ¶¶ 105–112. Dr. Michniak-Kohn further persuasively testifies that the combination of Native and Bianchi '254 teaches the "at least about 25% of liquid triglyceride" limitation. Dr. Michniak-Kohn states:

> Bianchi '254 describes an antiperspirant stick with 40-80% oil. In Bianchi '254, part of the oil carrier (20-40% of the total) is a natural oil such as a liquid triglyceride. Ex. 1008, ¶¶ [0017], [0050]. Bianchi '254 also describes that the antiperspirant stick further comprised 20-40% of silicone oil. *Id.*, ¶ [0016]. In my opinion, a POSA would have understood that silicone oil and liquid triglyceride both functioned as "carrier oils" that, among other things, solubilize and disperse other ingredients (*id.*, ¶¶ [0015], [0023], cl. 17, 18), and therefore would have been motivated to use even more liquid triglyceride when formulating a natural deodorant stick (like in Native) that did not include, e.g., silicone oil, to meet what the '915 Patent acknowledges was an existing consumer preference (Ex. 1001, 1:19-33). A POSA would have appreciated that one would want to consider the total oil composition as a guide.

> Thus, in my opinion, a POSA would have been motivated to use at least about 25% of a liquid triglyceride in a deodorant stick, and would have had a reasonable expectation of success in doing so, inasmuch as (i) Bianchi '254 already suggests the same with its 20-40% amount, (ii) a POSA would have, based on Bianchi '254's overall oil teachings, used even more of the liquid triglyceride when making a stick with only that oil carrier (as taught by Native).

Ex. 1002 ¶¶ 116–117. We agree with this assessment of the combination of the teachings of Bianchi '254 and Native. A POSA would have increased the amount of liquid triglyceride in Bianchi '254 in the 40 to 80% range for the total amount of oil disclosed in Bianchi '254, when applied to the Native formulation that is natural and does not contain any silicone oil, to arrive at the at least 25% by weight liquid triglyceride as recited in claim 1.

We also agree that this conclusion is supported by the overlapping ranges for oil disclosed in Bianchi '254 and claim 1. Ex. 1002 ¶ 118. Further, we agree with Dr. Michniak-Kohn that "there was nothing critical or unexpected about the claimed range relative to Bianchi '254's ranges. As a POSA would have understood, the minor deviations in the ranges yield predictable variations in degree and do not produce a different kind of result. For example, using more liquid triglyceride generally yields a softer stick, merely changing the degree of hardness." *Id.* ¶ 145. Patent Owner offered no rebuttal to this assessment that changing the amount of liquid triglyceride is merely a variation in degree and not result. *See KSR*, 550 U.S. at 416 (stating "combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results"). Moreover, we are not persuaded by the testimony of Dr. Lochhead to the contrary. *See* Ex. 2011 ¶ 106 ("The POSA would not have been motivated to consider the teachings of Bianchi-254 in combination with Native, as

Bianchi-254's prominent use of non-natural ingredients and aluminum would defeat the core objective of Native to provide aluminum-free, natural deodorant products.").

For at least the reasons discussed above, we determine that Petitioner has shown by a preponderance of the evidence that claim 1 would have been obvious over Native and Bianchi '254.

2. Dependent claim 9

Claim 9 depends from claim 1 and further requires the deodorant stick to comprise baking soda. Petitioner contends that Native's deodorant stick comprised baking soda, which is a known antimicrobial. Pet. 52.

Patent Owner contends that Petitioner's rationale for combining Native with Easy Homemade to substitute baking soda with magnesium hydroxide in Ground 6 is directly contrary to its argument here to use baking soda. PO Resp. 60. Patent Owner further contends that that "Petitioner makes no effort to reconcile its pro- and anti-baking soda arguments as to these claims and explain why a POSA would have been motivated to remove baking soda and then also decide to keep it in." PO Resp. 68 (citing Ex. 2011 ¶ 127–128)

Having considered the parties' positions and evidence of record, summarized above, we agree with Petitioner and are not persuaded by Patent Owner's arguments to the contrary. The specific set of advantages and disadvantages for using baking soda in a deodorant was known to a person of skill in the art. *See*, *e.g.*, Ex. 1007, 11 (Native's inclusion of baking soda); Ex. 1005, 132–133 (Lamb teaching that baking soda is an "excellent deodorant" that "lasts through the entire day."); Ex. 1017, 5–7 ("[T]he absence of baking soda means no itchy, red pits . . . for those of us with sensitive skin.”); Ex. 1009, 6, 9 (discussing the “rash you get after using deodorant with baking soda”); Ex. 1002 ¶ 150 (“A POSA would have known, however, that deodorants containing baking soda (Native’s antimicrobial) could cause skin irritation for some users.”). The inclusion or exclusion of baking soda presents no unexpected result and the evidence of record establishes that it would have been an obvious matter of design choice within the skill of the art to balance the advantages and disadvantages associated therewith. *See*, *e.g.*, *Medichem*, 437 F.3d at 1165

3. Dependent claim 15

Claims 15 depends from claim 1 and further requires the deodorant stick to have a hardness from about 80 mm*10 to about 140 mm*10, as measured by penetration with ASTM-D 1321 needle. Patent Owner contends that, “[j]ust like the Lesniak-based grounds, Petitioner cannot show its Native combination discloses the hardness range required by claim 15 of the ’915 Patent.” PO Resp. 51. Here, Patent Owner reiterates its argument that the ’915 patent discloses a hardness test that is distinct from the standard ASTM D-1321 test. *Id.* at 51–55. We have previously addressed this issue with regard to the Lesniak grounds and rely on our analysis for that ground here. We find this argument unpersuasive to the reasons provided above in the Lesniak grounds. *See* Section III.F.

4. Dependent claims 2, 5–8, and 11–14

Petitioner argues that claims 2, 5–8, and 11–14 would have been obvious over the combination of Native and Bianchi ’254. Pet. 54–61. Patent Owner presents the arguments discussed above regarding the limitations of independent claim 1 but does not present arguments or direct us to evidence against Petitioner's challenges that are specific to the limitations of dependent claims 2, 5–8, and 11–14. *See generally* PO Resp.

Upon review of Petitioner's undisputed argument, which we adopt as our own, and the evidence of record, we determine that Petitioner has demonstrated by a preponderance of the evidence that each of the limitations recited in claims 2, 5–8, and 11–14 are suggested by the above-identified prior art and that the skilled artisan would have had reason to make the combination with a reasonable expectation of success.

5. *Summary*

The preponderance of the evidence supports Petitioner's argument that claims 1–2, 5–9, and 11–15 would have been obvious over the combination of Native and Bianchi '254. Accordingly, we determine that claims 1–2, 5–9, and 11–15 are rendered obvious by the combination of Native and Bianchi '254.

H. Ground VI: Obviousness over Native, Bianchi '254, and Easy Homemade

Petitioner asserts that claims 3, 4, and 10 are unpatentable under 35 U.S.C. § 103 as obvious over the combination of Native, Bianchi '254, and Easy Homemade. These claims each depend directly from claim 1 and further require that the deodorant stick include the antimicrobial as a powder with a water solubility of at most about 90 g/L at 25° C (claim 3); is substantially free of baking soda (claim 4); or includes magnesium hydroxide in the antimicrobial (claim 10). Pet. 65; Ex. 1001, 16:15–17, 18–19, 30–31.

Easy Homemade discloses replacing baking soda with magnesium hydroxide as the antimicrobial. *Id.* (citing Ex. 1009, 6). Petitioner concludes "[a] POSA would have been motivated to substitute magnesium hydroxide powder for baking soda in the deodorant disclosed in Native for [this reason].” *Id.* Petitioner asserts this combination has a reasonable expectation of success because “a POSA would have appreciated that deodorant compositions could be formed through various combinations of known ingredients, such as antimicrobials, as simple design choices.” *Id.* at 49 (citing to Ex. 1002 ¶ 155).

Patent Owner charges that “Petitioner fails to adequately show why a POSA would have been motivated to look at the amateur Easy Homemade soap blog in the first place or why a POSA would have found it necessary to modify the already-successful Native product.” PO Resp. 58. But as Petitioner explained in its Petition, “Easy Homemade is a printed publication from a website dedicated to natural deodorant formulations.” Pet. 62. Since the Native product is a deodorant stick and Easy Homemade describes how to make deodorant compositions, a POSA would have reason to review the disclosure of Easy Homemade and apply its teachings, for example, its teaching that magnesium hydroxide could be used to replace baking soda as an antimicrobial in deodorant compositions. Pet. 63–64 (citing Ex. 1009, 6, 9); *see also* Ex. 1002 ¶¶ 152–154 (Dr. Michniak-Kohn testifying that Native was a well-known product, that baking soda was known to “cause skin irritation for some users,” and that “it would be obvious to use magnesium hydroxide as a substitute to baking soda in the composition of Native, as taught by Easy Homemade.”). Hence, contrary to Patent Owner’s contention that Petitioner has not articulated why a POSA would be motivated to combine Native, Bianchi ’254, and Easy Homemade, Petitioner has, indeed, articulated a persuasive motivation as to why Native, Bianchi ’254, and Easy Homemade would be combined.

Patent Owner also asserts that Petitioner misreads Easy Homemade as teaching the removal and substitution of baking soda, but "[r]ather, the Easy Homemade hobbyist tells her followers to purchase a pre-made baking soda-free deodorant base—a base Petitioner's expert could not identify the ingredients of—and add some magnesium hydroxide powder to it." PO Resp. 57. Patent Owner further contends that, "[a]s Petitioner's expert confirms, Easy Homemade 'doesn't talk about removing any ingredients from these sticks; she's talking about adding ingredients.'" *Id.* 57–58 (citing Ex. 1012, 121:1–9).

Petitioner responds that using prior art elements according to their established functions is obvious. Reply 24 (citing *KSR*, 550 U.S. at 417; *Intel Corp. v. PACT XPP Schweiz AG*, 61 F.4th 1373, 1380-81 (Fed. Cir. 2023)). In this case, as Petitioner explains,

> A known antimicrobial for deodorants, baking soda, caused irritation for some users, and another known antimicrobial for deodorants, magnesium hydroxide, ameliorated that issue when substituted for baking soda. Ex. 1002, 63-64; Ex. 1081, 53:6-55:12. "Nothing more is required to show a motivation to combine under *KSR*." *Intel*, 61 F.4th at 1381.

Reply 24.

Dr. Michniak-Kohn points to evidence that "a POSA would have known that deodorants containing baking soda could cause skin irritation for some users" and that "Easy Homemade confirms what a POSA would have known—namely, that magnesium hydroxide could be used to replace baking soda as an antimicrobial in deodorant compositions." Ex. 1002 ¶ 152 (citing Ex. 1009, 6, 9). Petitioner further notes that Easy Homemade expressly discusses substituting magnesium hydroxide for baking soda because Easy Homemade states that "baking soda can easily be swapped for magnesium hydroxide." Reply 24 (citing Ex. 1009, 6, 9). We agree with Petitioner that "Easy Homemade just directs the POSA to substitute one known antimicrobial for another." *Id.*

Finally, Patent Owner contends that Easy Homemade does not qualify as publicly accessible prior art because the evidence of record does not establish "that interested persons exercising reasonable diligence would have been able to locate Easy Homemade at the time of the priority date of the challenged patent." PO Resp. 56. We have considered this argument but determine that Petitioner has shown by a preponderance of the evidence that Easy Homemade was publicly accessible. A reference is considered publicly accessible if it is "made available to the extent that persons interested and ordinarily skilled in the subject matter or art, exercising reasonable diligence, can locate it." *Jazz Pharms., Inc. v. Amneal Pharms., LLC*, 895 F.3d 1347, 1355 (Fed. Cir. 2018). To "determine whether an interested researcher would have been sufficiently capable of finding the reference and examining its contents," we "must consider all of the facts and circumstances surrounding the disclosure." *In re Lister*, 583 F.3d 1307, 1312 (Fed. Cir. 2009); *see also In re Klopfenstein*, 380 F.3d 1345, 1350 (Fed. Cir. 2004) (explaining that a determination of whether a reference qualifies as a printed publication "involves a case-by-case inquiry into the facts and circumstances surrounding the reference's disclosure to members of the public").

Patent Owner's main argument appears to be that Soap Deli News was not indexed. *See* PO Resp. 61. As Petitioner points out, however, such evidence is not necessarily required in all cases. Reply 25 (citing *Voter Verified, Inc. v. Premier Election Soln's*, 698 F.3d 1374, 1380 (Fed. Cir. 2012)). "Rather, absent evidence of indexing, testimony indicating that the particular online publication or website on which the reference was published was well-known to the community interested in subject matter of the reference, and that, upon accessing the website, those interested would have found the reference using the website's own search functions can support the ultimate determination that a given reference was publicly accessible." *Sandoz Inc. v. Abbvie Biotechnology LTD.*, IPR2018-00156, Paper 11, *11–12 (PTAB June 5, 2018) (citing *Voter Verified*, 698 F.3d at 1381).

Petitioner presents evidence from the Wayback Machine for the Easy Homemade Blogpost, *see* Ex. 1009, 1–2, in addition to three other Soap Deli News posts, *see* Exs. 1017–1019. Dr. Mishniak-Kohn testifies "Easy Homemade was widely accessible, being part of a series of related 'Soap Deli News' articles dedicated to deodorant formulations with significant engagement from individuals globally." Ex. 1002 ¶ 151 (citing Ex. 1009; Exs. 1016–1018). Dr. Mishniak-Kohn also testifies that "a POSA would have appreciated that such websites from independent formulators striving for more natural options present an indication of consumer preferences, which a POSA would desire to meet in formulating new commercial products." *Id.* We credit Dr. Mishniak-Kohn's testimony here, and determine that Easy Homemade was publicly accessible and qualifies as a printed publication.

Based on the full record before us, we determine that Petitioner has shown by a preponderance of the evidence that claims 3, 4, and 10 would have been obvious over Native, Bianchi '254, and Easy Homemade.

IV. CONCLUSION[4]

Based on the fully developed trial record, Petitioner has demonstrated by a preponderance of the evidence that claims 1–15 of the ’915 patent are unpatentable.

In summary:

Claim(s)	35 U.S.C. §	Reference(s)/ Basis	Claim(s) Shown Unpatentable	Claim(s) Not Shown Unpatentable
1, 3–8, 11–14	102/103	Lesniak	1, 3–8, 11–14	
2, 10	103	Lesniak, Phinney	2, 10	
9	103	Lesniak, Lamb	9	
15	103	Lesniak, Bianchi ’518,	15	
1–2, 5–9, 11–15	103	Native, Bianchi ’254	1–2, 5–9, 11–15	
3, 4, 10	103	Native, Bianchi ’254, Easy Homemade	3, 4, 10	
Overall Outcome			1–15	

V. ORDER

In consideration of the foregoing, it is

[4] Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner’s attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. *See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2).

ORDERED that claims 1–15 of the '915 patent have been shown to be unpatentable; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to this proceeding seeking judicial review of our decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

FOR PETITIONER:

Justin Oliver
Sarah Brooks
VENABLE LLP
joliver@venable.com
ssbrooks@venable.com

FOR PATENT OWNER:

David Maiorana
Marlee Hartenstein
Jesse Wynn
JONES DAY
dmaiorana@jonesday.com
mhartenstein@jonesday.com
jwynn@jonesday.com